

Chapter 4

Impact of pressure and charge variations on EEL spectra of embedded xenon gas clusters.

This chapter explores the changes in the EELS features stemmed from the structural modifications of Xenon (Xe) gas clusters encapsulated within graphene. The study examines two specific configurations: clusters composed of five Xe atoms and those containing two Xe atoms. Through DFT-EELS simulations, variations in charge and pressure are systematically introduced to assess their impact on the structural changes within the system and the corresponding EELS spectra.

4.1 Xenon gas clusters: insights and the drive behind this work

Xe is a fascinating noble gas due to its unique properties and diverse applications across critical fields such as research, industry, and medicine [1–3]. In nuclear applications, Xe is both a fission byproduct and a crucial element in nuclear waste management. Innovative approaches, such as trapping xenon in silica nanocages, have been explored for nuclear waste remediation and the nonproliferation of nuclear weapons [4]. Its medical applications include its use as a safe and effective anaesthetic and in medical imaging [5], further demonstrating its inertness and versatility. In dark matter detection, liquid xenon is highly sensitive and has a low energy detection threshold, making it well-suited for detecting weakly interacting massive particles [6]. Xe clusters in helium nanodroplets exhibit unique photoionisation dynamics, where direct inner-shell photoionisation dominates over charge transfer ionisation, primarily forming singly charged Xe ions despite the possibility of higher charge states through Auger decay [7]. Under femtosecond laser excitation, electron emissions increase significantly at low energies due to a two-step ionisation process involving helium ionisation by Xe seed electrons [8]. The ionisation process also shows a steplike intensity threshold for highly charged ions at unexpectedly low laser intensities [9]. Additionally, helium helps xenon clusters reach higher charge levels when hit by strong laser light, without changing their usual electron behaviour. [10].

Xe gas clusters embedded in layers or shells of other materials are achieved using ultralow-energy ion irradiation, where noble gas ions penetrate the first shell but are stopped by the second shell. This confinement creates a "sandwich" structure that stabilises the Xe atoms, preventing their escape and enabling their study at room temperature or higher. Xe gas clusters are gaining significant interest in condensed matter physics and industrial usage. From an industrial perspective, shell-like structures with Xe nano-traps in porous materials exhibit remarkable selectivity and adsorption capabilities, making them highly efficient for separating Xe from other gases like Krypton, a critical process in industries requiring pure gas streams [11]. Furthermore, in plasma discharge technologies, the incorporation of Xe

clusters into flexible polymer matrices has been shown to enhance luminescence, optimise luminance conditions, and reduce operational voltages, paving the way for more efficient and sustainable lighting solutions [12]. These exceptional properties and applications underscore the importance of Xe and make it a gas of significant scientific and practical interest.

In condensed matter physics, embedding Xe clusters in materials like graphene enables detailed structural studies at elevated temperatures, providing opportunities for direct imaging and in-depth analysis of atomic structures and dynamics [13]. In this study, we perform DFT-EELS simulations to investigate two xenon gas clusters embedded within graphene layers, aiming to interpret the effects of charge and pressure observed in experimental EELS data. On the basis of experimental data, we construct the cluster models, incorporate realistic conditions by introducing charge and pressure in the system, and analyse the resulting changes in the simulated spectra. Finally, we compare these results with experimental observations to draw meaningful correlations.

4.2 Experimental observations

The samples were prepared by using ultralow energy ion irradiation technique [13] by the group of Physics of Nanostructured Materials at the University of Vienna. In this process Xe atoms were implanted between graphene sheets. The graphene sheets (exfoliated and chemically vapourised) were irradiated with singly charged ultralow energy (55-65 eV) Xe ions to introduce Xe inside the graphene bilayers. The samples were then observed in a Nion UltraSTEM 100 microscope. The scanning transmission electron microscope - annular dark field (STEM-ADF) images are shown in Fig. 4.1. Xe clusters in multiple groups can be spotted inside a single image. The clusters have no particular order nor a periodicity. For a better view, a collection of multiple Xe clusters collected throughout the sample is provided in Fig. 4.2. Further, EELS measurements were carried out for two clusters (Xe_2 , Xe_5) and are shown in Fig. 4.3. For both the cases, the EELS spectra exhibit distinct features corresponding to the $M_{4,5}$ edges of Xe, thereby confirming its presence as isolated gas clusters.

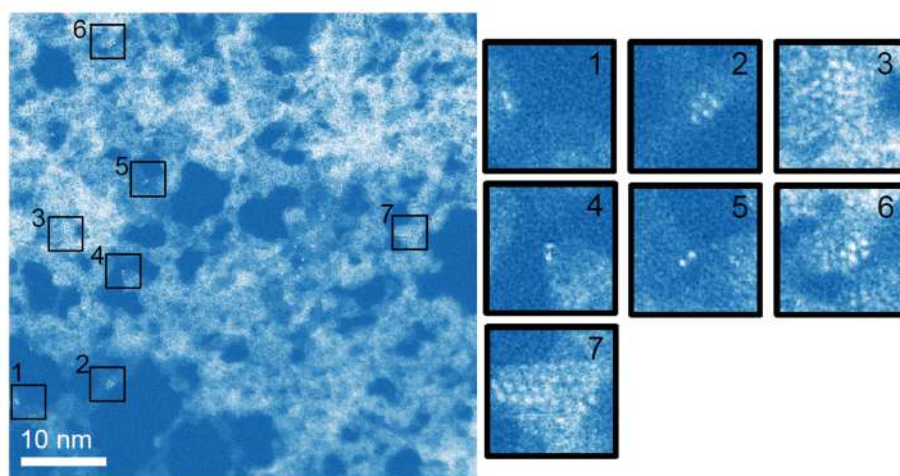


Fig. 4.1 Double Gaussian Fourier filtered STEM-ADF depicting the spread of Xe gas clusters across the sample [13].

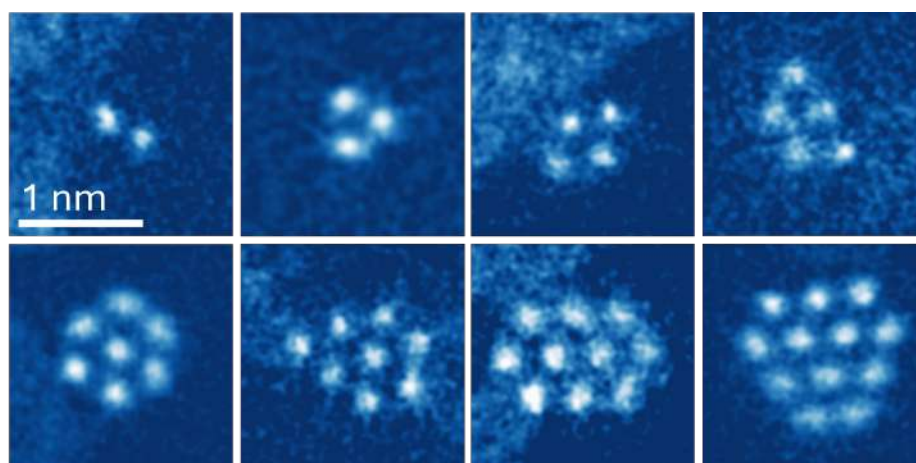


Fig. 4.2 Double Gaussian Fourier filtered STEM-ADF images of Xe gas clusters from 2 to 12 atoms [13].

Due to the presence of graphene around the Xe atoms, the Xe clusters form a 2D shape and are exposed to pressure from the surrounding graphene sheets [13]. Further, the gas clusters, may have charged behaviour as well due to electron beam irradiation during EELS. Both the charge and pressure may have an effect on the EELS of the Xe gas clusters. In order to interpret the effect of these parameters on the experimental EELS results, EELS simulations through DFT were carried out.

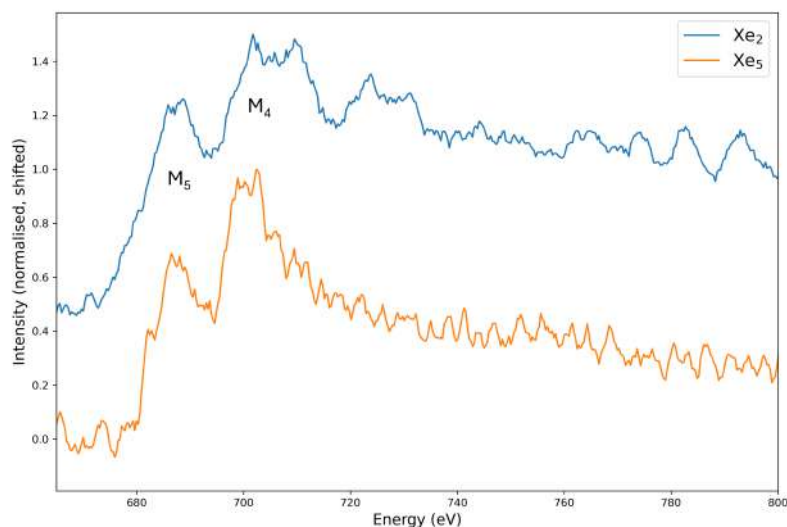


Fig. 4.3 EELS spectra for Xe_5 and Xe_2 gas clusters, depicting the $M_{4,5}$ edges.

4.3 EELS simulation of Xe gas clusters

Among the various Xe clusters observed in the sample, two clusters containing 2 (Xe_2) and 5 (Xe_5) Xe atoms were selected for EELS simulations. The selection criterion was primarily set by the significant size of Xe atoms, which requires high computational demands. For the simulations, the two-dimensional structures of the clusters were first modelled and structurally relaxed using GPAW [14]. EELS simulations were subsequently performed on the relaxed configurations using WIEN2k [15]. Additionally, further simulations were conducted to investigate the effects of introducing charge and applying pressure to the clusters, providing deeper insights into their EELS signatures.

4.3.1 Simulation structures

For the Xe_2 and Xe_5 , the initial structures were designed based on interatomic distances derived from the experimental images shown in Fig. 4.2. These preliminary structures were then relaxed using the GPAW software [14], employing the Lennard-Jones potential with a cutoff criterion of $0.001 \text{ eV/\AA}$. To simulate the effects of pressure on the system, the

relaxed structures were subjected to external forces of 0.2, 0.4, and 0.8 eV/Å. Following the application of each force, the structures were re-relaxed to account for the influence of pressure. The resulting relaxed configurations, including those under pressure, are presented in Fig. 4.4. The results indicate that increasing pressure causes the Xe atoms to move closer together, leading to a reduction in the interatomic distances.

4.3.2 EELS simulation

EELS simulations were performed on each relaxed structure using WIEN2K [15]. The self-consistent field (SCF) calculations were performed with the energy and charge convergence criteria set to 0.0001 eV and 0.001 e, respectively. Given the non periodic nature of the system, the number of k-points was set at 4, and the RKMax value was set at 6.5. The Perdew–Burke–Ernzerhof (PBE) functional was utilised for the simulations. Further, to study the effect of charge, a charge of +1 and +2 was put into the system by removing one and two electrons from the respective cluster.

4.3.3 EELS simulation results

The EELS simulations were performed individually for each atom in the system, and subsequently, the average spectrum for all Xe atoms was calculated to provide a representative view of the system's electronic structure. The results of these simulations for Xe₂ and Xe₅, accounting for the effects of charge states and applied pressure, are presented in Fig. 4.5 and Fig. 4.6, respectively.

For the study of simulated EELS spectra, changes in the ELNES of the M_{4,5} edges have been selected for observation. In the simulated spectra, it exhibits distinct spectral modifications that correlate with variations in pressure and charge state. For all the charge states, as the pressure increases, the M_{4,5} edges become sharper with further fine structures being visible as well. This indicates that the application of pressure significantly influences the local electronic environment of Xe atoms, possibly through changes in bond lengths, coordination, or local field effects.

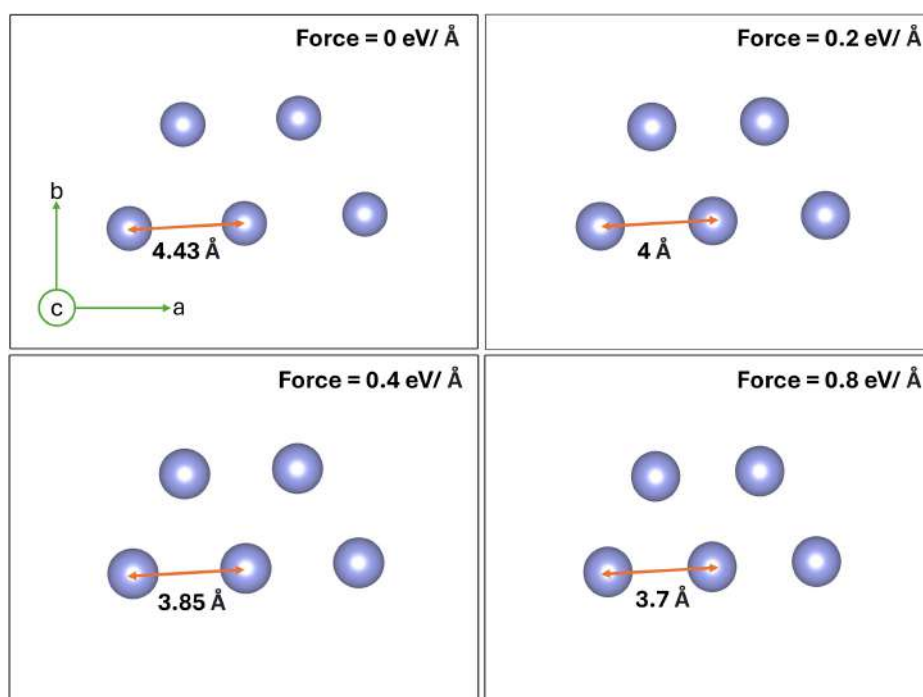
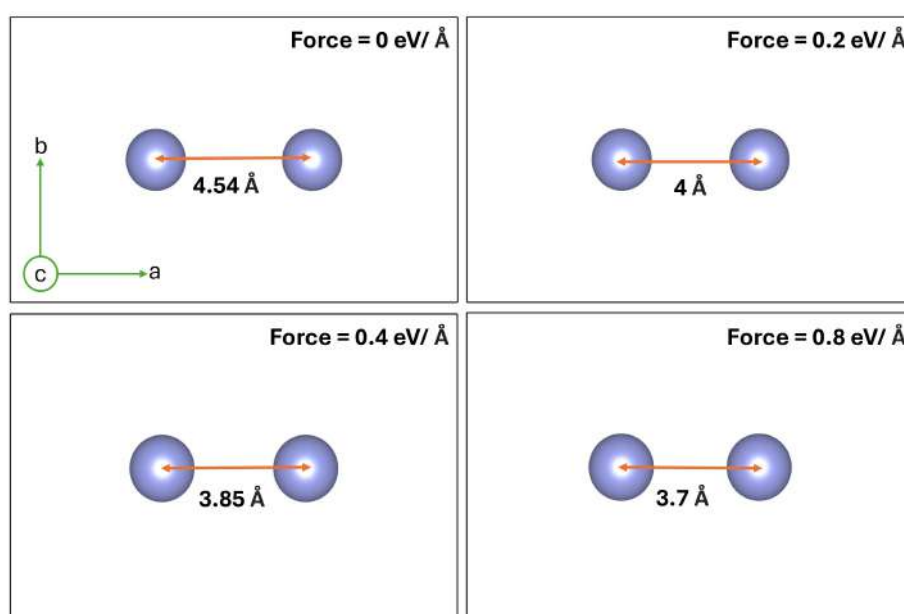
(a) Xe₅ cluster.(b) Xe₂ cluster.

Fig. 4.4 Xe cluster structures with force variations and the corresponding interatomic distance after relaxation of the structure.

A similar yet more prominent effect can be observed when higher charge states are introduced to the system. The $M_{4,5}$ edges along with their fine structures, can be seen evolving as the charges increase. This suggests that the electronic perturbations introduced by charging further enhance the sensitivity of the system to applied pressure. The pronounced splitting in higher charge states could be attributed to increased electronic repulsion or redistribution of the electron density within the system, which modifies the excitation probabilities in the EELS process.

The interplay between charge and pressure introduces a complex dependency, where higher charge states and increased pressure introduces spectral changes. These observations highlight the sensitivity of EELS to subtle changes in the local environment and opens the door to valuable insights into the electronic and structural characteristics of Xe clusters under varying external conditions.

4.4 Comparison of EELS simulation with experimental data

The comparison of experimental and the simulated EELS data are presented in Fig. 4.7 and Fig. 4.8 for Xe_2 and Xe_5 , respectively. Only the pressure states of 0.4 and 0.8 eV/Å have been considered for the comparison, as the states of 0 and 0.2 eV/Å clearly show relatively less features, as observed in Fig. 4.5 and 4.6. EELS simulations performed using WIEN2K often need a recalibration of the energy axis due to various factors, including relativistic calculations, broadening effects, and initial parameters [16–19]. Consequently, to improve the fit, the simulated spectra have been slightly shifted by about 2 and 6 eV towards lower energy for Xe_2 and Xe_5 respectively.

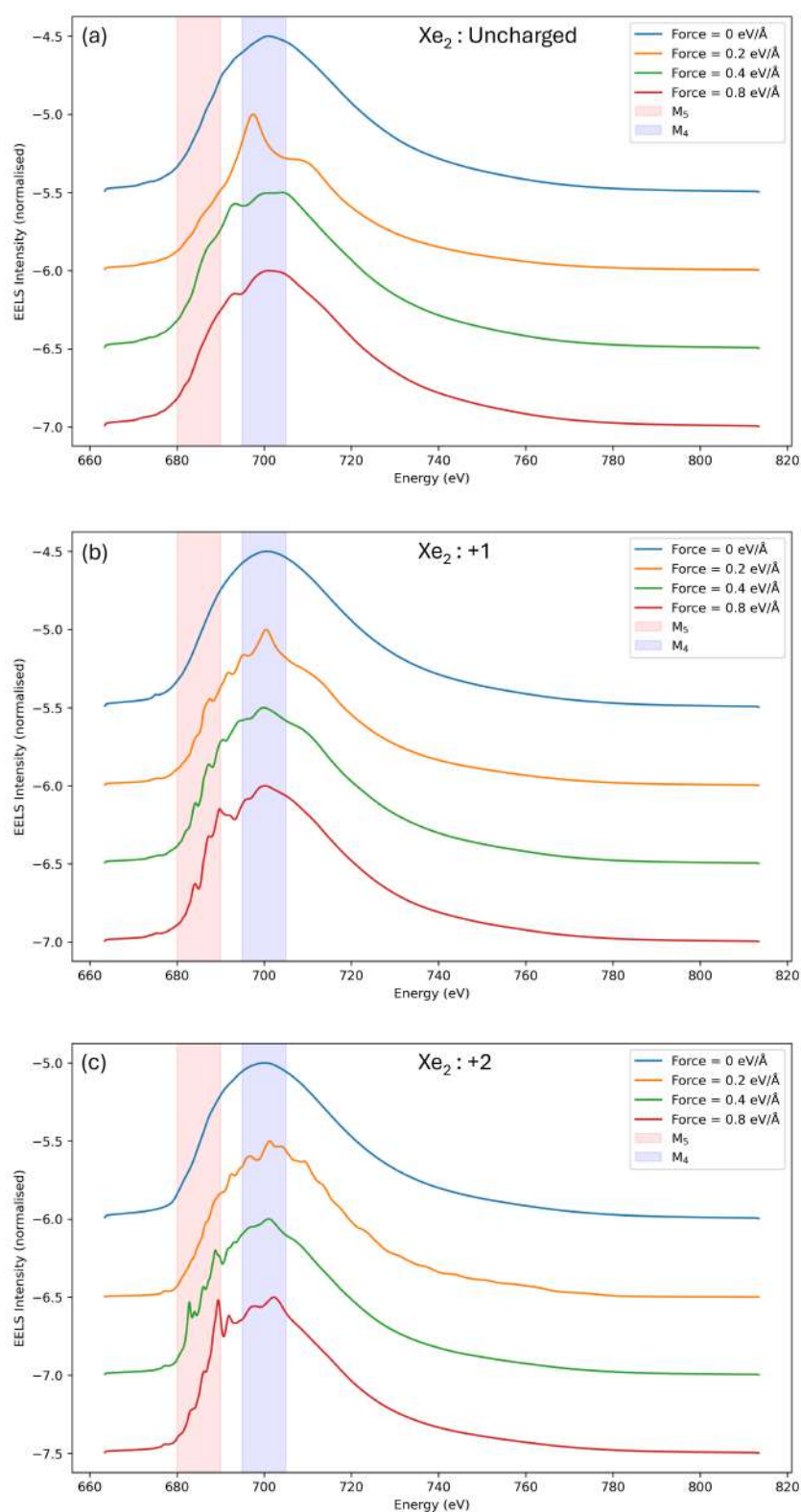


Fig. 4.5 Xe₂ EELS simulation with charge (a) 0 (b) +1 (c) +2 and four pressure (force) variations. The introduction of charge and pressure introduce changes in the M_{4,5} region.

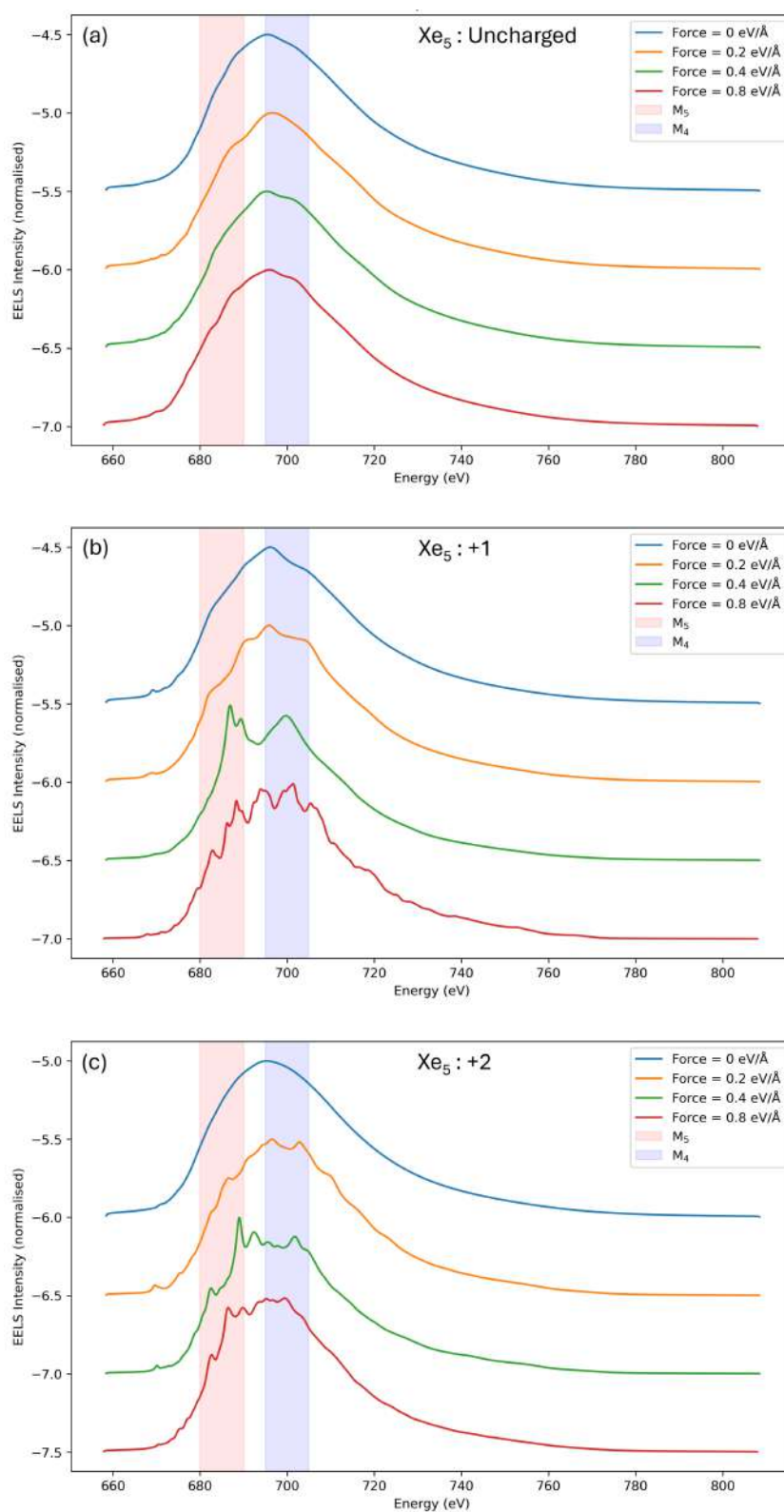


Fig. 4.6 Xe_5 EELS simulation with charge (a) 0 (b) +1 (c) +2 and four pressure (force) variations. The introduction of charge and pressure introduces changes in the $M_{4,5}$ region.

For a better comparison, the relevant features (named as A, B, C, D, and E on the plots), along with the relative peak heights in the M_5 and M_4 energy regions have been considered. For both the cases of Xe_2 and Xe_5 , besides the main edges, more features can be seen in the +2 charge state than in the +1 charge state.

For Xe_2 , the comparison between the experimental and simulated EELS data for both force values of 0.4 and 0.8 $\text{eV}/\text{\AA}$ in the +1 & +2 is in relative good agreement with the experimental EELS data. It can be observed that for both the pressures, in the +1 and +2 state, features A, B, and C show a good match. However, features D and E align more closely with the 0.8 $\text{eV}/\text{\AA}$ with +2 charge than the 0.4 $\text{eV}/\text{\AA}$ with +2 charge.

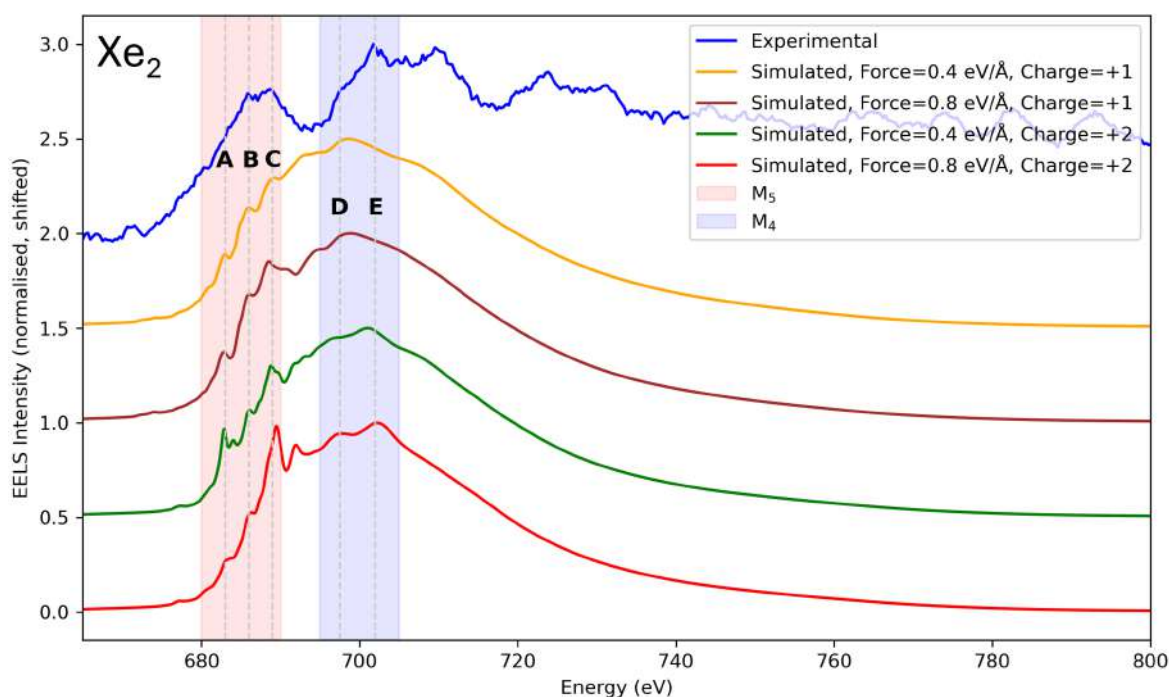


Fig. 4.7 Xe_2 comparison of experimental and simulated EELS data. Spectral features (A, B, C, D, and E) have been marked for an easier comparison between the plots.

Similarly, for Xe_5 , in the M_5 region, the simulated EELS spectra for both force values of 0.4 and 0.8 $\text{eV}/\text{\AA}$ in the +1 and +2 charge states tend to fit the experimental data. For the 0.4 $\text{eV}/\text{\AA}$ case, only two features (B, C) for +1 charge and (A, C) for +2 charge match the experimental data, whereas for 0.8 $\text{eV}/\text{\AA}$, all three features (A, B, and C) for the +2 charge

match the experimental data. Furthermore, the relative heights between the M_4 and M_5 edges correspond more closely with the $0.8 \text{ eV/\AA}$, $+2$ case. All these observations suggest that the simulated data for $0.8 \text{ eV/\AA}$ with a $+2$ charge state provide closer agreement with the experimental data than the others. In the M_4 region, both experimental and simulated data exhibit fine structures (D, E). However, a direct and precise correspondence between them is difficult to establish due to subtle variations. For example, feature D in the experimental data is a good match for the $0.8 \text{ eV/\AA}$ with $+1$ and $+2$ charge states as well as the $0.4 \text{ eV/\AA}$ with $+1$ charge but none of them is a good fit for the feature E. Similarly, feature E in the experimental data aligns well with the $0.4 \text{ eV/\AA}$ with $+2$ charge but the feature D in the same case does not.

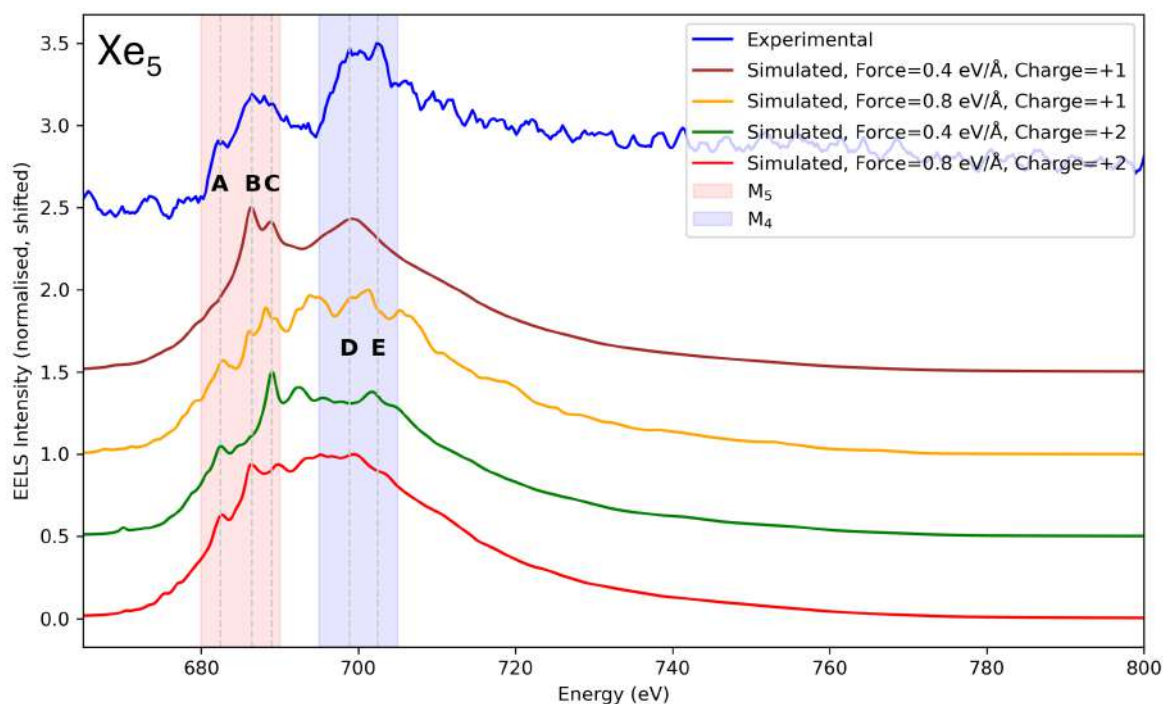


Fig. 4.8 Xe_5 comparison of experimental and simulated EELS data. Spectral features (A, B, C, D, and E) have been marked for an easier comparison between the plots.

It is worth noting that the minor discrepancies observed in the simulations highlight the intricate nature of the system and suggest that, while the simulations give us an idea about the effect of pressure and charge on the EELS, further refinements could enhance the accuracy

of relative peak intensities and ELNES. In particular, incorporating the effects of graphene, though computationally demanding, presents an exciting avenue for future improvements. Additionally, the $M_{4,5}$ edge fine structure is less striking in the simulations, which can be attributed to the inherent design of WIEN2K, as it is primarily optimised for solids [15]. In bulk materials, the presence of a larger number of atoms leads to stronger electronic interactions, naturally enhancing spectral features such as peak splitting. However, in our case, the relatively small number of atoms in the confined Xe clusters results in weaker interactions, making the splitting less prominent. Despite this, the overall trends remain consistent, and further refinements in computational methods could provide even greater accuracy in modelling such clusters.

Overall, by combining DFT-EELS simulations with experimental data, we have explored how external influences, such as confinement effects (pressure, charge) modulate the spectral features of Xe clusters.

4.5 Discussion

This chapter illustrates the fruitful use of DFT-EELS simulations to investigate electronic structure changes of confined noble gas clusters. Systematic investigation of pressure-induced effects in Xe clusters indicates that confinement distinctly alters the local electronic surroundings, as shown by the development of $M_{4,5}$ edge fine structure with the increased applied pressure. The fact that spectral properties become more defined and sharp at higher pressure implies that atomic compression alters the density of states at the Fermi level.

These observations are important for explaining how two-dimensional confinement between graphene sheets influences the intrinsic properties of noble gas atoms, which are otherwise considered chemically inert. The fact that the charge states provide the best match with experiment indicates that the electron beam irradiation carried out during EELS measurements largely influences the charge configuration in these small clusters.

The good correlation of DFT-EELS simulations and experimental findings justifies the computational method in the investigation of such trapped atomic arrangements. The

consistency of simulated and experimental spectral characteristics proves that first-principles calculations can be used to help predict the electronic response of matter in non-standard conditions.

These findings add to the general understanding of how confinement in dimensions impacts atomic properties. The capacity to alter the electronic structure in noble gases using geometric restriction provides prospects for creating new functional materials in which inert elements may have vital functions.

4.6 Conclusions

Overall, the findings of the study can be summarised as the following.

- This study investigated the impact of pressure (0.2, 0.4, and 0.8 eV/Å) and charge (0, +1, and +2) on the EELS spectra of embedded Xe₂ and Xe₅ gas clusters using DFT-EELS simulations and experimental comparisons.
- STEM-ADF imaging and EELS measurements confirmed the presence of Xe clusters inside graphene layers, providing a reference for theoretical simulations.
- Structural relaxation of the clusters was performed using GPAW, and EELS spectra were simulated using WIEN2K, incorporating charge and pressure variations to replicate experimental conditions.
- Increasing pressure (0.4 and 0.8 eV/Å) led to a splitting of the M_{4,5} edges, demonstrating a significant impact on the local electronic structure of Xe atoms.
- Higher charge states (+1 and +2) modified the spectral features further, with +2 charge at 0.4 – 0.8 eV/Å showing the best agreement with experimental data for both Xe₂ and Xe₅ clusters.
- With the increase in computational facilities, further refinements in the simulations can increase the accuracy, thereby enabling a deeper understanding of the physics of noble gas clusters.

4.7 References

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